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PRN: B26ET2024 Div: SY3 Class: C3

INPUT:

```
#include<stdio.h>
#include<stdlib.h>
struct node // node defining for tree
{
struct node* left;
struct node* right;
int data;
};
struct stack // node defining for stack
{
struct node* data;
struct stack* next;
};

void push(struct stack** top,struct node* n); //function declation
struct node* pop(struct stack** top);
int isEmpty(struct stack* top);

int Inorder(struct node* root) //Inorder Traversing function
{
struct node* temp = root;
struct stack* s_temp = NULL;
int flag = 1;
while(flag) //Loop run untill temp is null and stack is empty
{
if(temp){
push(&s_temp,temp);
temp = temp->left;
}
else{
if(!isEmpty(s_temp)){
temp = pop(&s_temp);
printf("%d ",temp->data);
temp = temp->right;
```

```
}
else
flag = 0;
}
}
}
void push(struct stack** top,struct node* n) //push node in stack
{
struct stack* new_n = (struct stack*)malloc(sizeof(struct stack));
new_n->data = n;
new_n->next = (*top);
(*top) = new_n;
}
int isEmpty(struct stack* top) // check if stack is empty
{
if(top==NULL)
return 1;
else
return 0;
}
struct node* pop(struct stack** top_n) // pop the node from stack
{
struct node* item;
struct stack* top;
top = *top_n;
item = top->data;
*top_n = top->next;
free(top);
return item;
}
struct node* create_node(int data) // create a node for tree
{
struct node* new_n = (struct node*)malloc(sizeof(struct node));
new_n->data = data;
new_n->left = NULL;
new_n->right = NULL;
return (new_n);
}

int main()
```

```
{
struct node* root;
root = create_node(8);
root->left = create_node(5);
root->right = create_node(4);
root->left->left = create_node(7);
root->left->right = create_node(6);
Inorder(root);
return 0;
}
```

OUTPUT:

```
PS D:\MMCOE\DSA\codes> D:\MMCOE\DSA\codes\practical.exe ""
7 5 6 8 4
```